

Final Year Research Project Proposal

on

AI Legal Intent Rewriter

by

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1 Project Overview

The focus of this project is on developing an AI-based solution capable of converting complex legal terms into simpler and more understandable language for common users. The system aims to help individuals understand what legal clauses mean, what obligations they impose, their limitations, and the potential risks involved. By analyzing and interpreting legal language, the solution will present explanations in clear and concise segments.

2 Background and Motivation

Legal documents are often written in highly technical and dense language, making them difficult for non-experts to interpret. Many individuals unknowingly agree to terms without fully understanding the implications. Although tools exist for drafting or researching legal documents, very few focus on simplifying and clarifying legal intent.

With advancements in natural language processing and transformer-based models, it is now feasible to analyze and summarize complex legal text. This project is motivated by the need to empower individuals and small businesses to better understand legal agreements without requiring immediate professional assistance.

3 Objectives

The key objectives of the project are:

1. Collect and preprocess datasets containing legal clauses.
2. Segment legal documents into individual clauses.
3. Classify clauses based on their legal intent.
4. Extract key entities such as dates, monetary values, rights, and obligations.
5. Generate simplified interpretations of legal clauses.
6. Identify potentially risky or critical clauses.
7. Develop a user-friendly web interface for document upload and result visualization.

4 Proposed Methodology

The project follows a multi-stage processing pipeline. Initially, legal documents are uploaded in PDF or text format. For scanned documents, an OCR module converts images into machine-readable text. The extracted text is then segmented into clauses.

A transformer-based model is used to classify each clause. Named Entity Recognition (NER) techniques extract critical information such as dates, amounts, and responsibilities. A rewriting module then generates simplified explanations for each clause. Additionally, a rule-based system flags clauses that may pose legal risks. Finally, a web application presents the processed output to users.

5 Tools to be Used

- **Programming Languages:** Python, JavaScript
- **Frameworks & Libraries:** TensorFlow, Hugging Face Transformers, spaCy
- **Database:** PostgreSQL, Redis
- **OCR Tools:** Tesseract OCR
- **Development Platform:** Windows/Linux with Docker
- **Hardware:** Standard laptop (GPU optional)

6 Expected Outcomes

The expected deliverables of the project are:

- A system capable of simplifying legal clauses.
- Structured explanations highlighting rights, obligations, and risks.
- Automatic identification of potentially critical clauses.
- A web-based dashboard for document upload and interpretation.
- Performance evaluation and scope for future enhancements.

7 Timeline

Project Phase	Duration
Literature Review and Dataset Search	Week 1 – Week 3
OCR Pipeline and Preprocessing	Week 4 – Week 6
Clause Segmentation and Classification	Week 7 – Week 10
Intent Rewriting Module	Week 11 – Week 12
UI Development and Risk Detection	Week 13 – Week 14
Testing and Improvements	Week 15 – Week 16
Documentation and Final Submission	Week 17

8 Team Roles

- **Swagat Sourav Mallick:** Dataset collection and OCR configuration.
- **Shnehal Mishra:** Clause classification and entity extraction.
- **Pratyush Mohanty:** Intent rewriting and risk detection.
- **Ashreeta Patra:** Frontend development and backend integration.
- **All Members:** Documentation, testing, presentation, and debugging.

9 Contact Details of Team Lead

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10 References

- Smith, J. & Patel, R. (2022). *Artificial Intelligence in Legal Systems*.
- Nguyen, T. (2021). Legal Document Analysis using NLP.

Student Signatures

We hereby declare that the above project proposal is prepared by us and all information provided is true to the best of our knowledge.

(Swagat Sourav Mallick)

(Shnehal Mishra)

(Pratyush Mohanty)

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(Office use only)

Assigned Supervisor: _____

Comments

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Project Status

Approved

Not Approved

(Dr. Rangaballav Pradhan)

(Gyana Ranjan Patra)

(Project Coordinators)

Date: _____

Place: _____